

NMSWorks: Building a solid foundation for seamless telecommunic ations around the world

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ABOUT NMSWORKS



NMSWorks migrates network operations to Google Cloud to support a flexible environment, monitor and fetch fault and performance data from subsea and terrestrial cables and alert customers with real-time data.

About NMSWorks

NMSWorks Software

(<https://www.nmsworks.co.in/>) is a Software Product and Services company specializing in developing Carrier Grade Operations Support Systems (OSS) and Business Support Systems (BSS) solutions. Its flagship solution CygNet monitors the networks and proactively ensures system and service availability to customers. Company HQ is situated in IIT Madras Research Park, Chennai, IND.

Industries: Telecommunications

Google Cloud results

- Calculates load data in just 3 minutes instead of 10+ days with accuracy
- Ensures uncompromised security with Cloud Armor and Cloud Load Balancing
- Creates custom models for prediction easily with AutoML
- Enables automated cross-border communication with Dialogflow

In the highly connected world we live in, it's easy to forget that deep under the sea, there are highly connected cables that form the backbone of the technology we rely on so heavily.

Working with some of the largest telecommunications companies, NMSWorks monitors and fetches

Red client onboarding from 10 to 8

Location: India

(<https://cloud.google.com/bigquery/>), [Cloud](#)



About Cloudside

Cloudside is a trusted team of cloud-native and data problem solvers, helping its customers drive the best business outcomes with the cloud as a catalyst. It brings decades of collective experience in cloud computing, data, and app engineering. Its mission is to bring this expertise and make it available for every customer to help solve complex cloud and data problems.

Google Cloud (<https://cloud.google.com/>)

Cloud AutoML (<https://cloud.google.com/automl/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Load Balancing
(<https://cloud.google.com/load-balancing/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Compute Engine
(<https://cloud.google.com/compute/>)

Dialogflow (<https://cloud.google.com/dialogflow>)

customers drive the best business

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Virtual Private Cloud (<https://cloud.google.com/vpc>)

performance data from subsea cables globally, sending automated information to customers with its software, CygNet. "We are a vendor company that provides the technical teleco's with inventory, fault, and explains Dr. Madanagopal Ramani Technical Architect at NMSWorks.

Google Chat

(<https://workspace.google.com/products/chat/>)

Google Drive

(<https://workspace.google.com/products/drive/>)

NMSWorks uses Compute Engine

(<https://cloud.google.com/compute>) to create its VMs, and to keep its networks secure, the company

leverages Virtual Private Cloud

(<https://cloud.google.com/vpc>) (VPC) and Cloud Load Balancing (<https://cloud.google.com/load-balancing>). It

uses Cloud SQL (<https://cloud.google.com/sql>) for databases, and BigQuery

(<https://cloud.google.com/bigquery/>) for data collection, storage, and analysis.

The team works with partner agency Cloudside

(<https://thecloudside.com/>), who operate as an extension of their in-house team, sharing best practices on deploying their applications stack on Google Cloud (<https://cloud.google.com/>) through support in deployment design considerations.

"With Google
Cloud, we
templatized the

environment,
making it easier
for us to create a
replica of the
same whenever
required
according to
project needs,
which saved us
a lot of time,
effort, and
resources."

*—Dr. Madanagopal
Ramchandran, Senior
Technical Architect at
NMSWorks*

Transitioning from on-prem to the cloud

In its earlier days, NMSWorks relied heavily on on-premise architecture to deploy its software.

"Onboarding new clients was a big task as we had to send a team to the customer's location, which could be in different countries," explains Dr. Madanagopal, adding that it would take between 15 to 25 days for installation to take place, depending on the project size, as every server had to be installed manually. Even after the setup was complete, the team had to allocate time and resources to manage the new

deployments on site, since there would be teething issues that needed to be resolved.

However, since working with Google Cloud in 2018, the deployment process has become much quicker. Now, a single setup can be done in just eight hours. "With Google Cloud, we templated the environment, making it easier for us to create a replica of the same whenever required according to project needs, which saved us a lot of time, effort, and resources," says Dr. Madanagopal. With the team managing global subsea networks and submarine cables across the Atlantic, the Pacific, and the South China Sea, being able to have full visibility and control on Google Cloud has transformed operations at the company.

Even when the pandemic happened in 2019 and made remote work the only option, NMSWorks found Google Cloud to be their "saving grace," as their daily operations were not affected. Since everything was in the cloud, NMSWorks' software continued to run unhampered. Furthermore, Cloudside's assistance in optimizing their cloud services with a cost-centric architecture approach ensured that NMSWorks always had the most optimum solutions on hand.

Processing large amounts of data seamlessly with BigQuery and MySQL

Submarine fiber optic cables can be very costly to repair, which is why it is important to ensure that the position of the cable is at the most optimum

location when laying it out. For this reason, it is important to calculate the density base and ensure that it does not interfere with the fishing areas. "We need to run calculations before laying submarine cables and decide where it runs through, and these calculations need to be very precise," says Dr. Madanagopal.

With around 10,000 messages sent per second, the calculations amounted to about 500 million records per day. This huge amount of data meant that calculating the outcomes required a platform that could support the load.

Initially, the team used MySQL to run the calculations. "After about 10 days of waiting for the calculations to conclude, we were still not able to come to a conclusive point," explains Dr. Madanagopal. The other option was to split the globe cells into smaller tables, which would take a lot of time and effort. "Eventually, we decided to go with BigQuery, which helped us a lot because that same job was completed in just three minutes, and we got the data and information we needed."

The team also used [AutoML](https://cloud.google.com/automl) (<https://cloud.google.com/automl>) to generate predictions. "Back then when we first started testing the capabilities of AutoML, I didn't know much about machine learning. But I had some 2D data. So I uploaded the CSV and input some information on what I needed to predict. In just a few hours, the model was ready," Dr. Madanagopal shares, adding that he was surprised at how simple the entire process was. "It was amazing how quickly we could

create custom models with AutoML. Even a layman with no technical expertise could do it."

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—Dr. Madanagopal
Ramchandran, Senior

Alerting customers in real time to solve critical issues

NMSWorks deploys the Cisco Cloud Services Router 1000V (CSR 1000V) in Google Cloud, which is connected to the submarine equipment for monitoring. If any faulty data is detected, an automatic alert would be sent through a phone or a chatbot to customers using Google API, so that they can react quickly to solve the problem.

For the chatbot to be effective in communicating with customers from around the world, NMSWorks leverages the capabilities of Dialogflow (<https://cloud.google.com/dialogflow>). "Integrating Dialogflow with Google Chat (https://workspace.google.com/intl/en_sg/products/chat/) , we are able to receive what customers type, and through natural language processing (NLP) technology, we can understand the intent and parameters of what is being asked. We would then pull the relevant data and send it back to the customer in their prescribed format," explains Dr. Madanagopal.

Catering to customer's needs while ensuring a secure gateway

While NMSWorks strongly encourages their customers to use Google Cloud, they are also able to

support those who may not want to log into the Google Cloud console. Using Cloud Monitoring API (<https://cloud.google.com/monitoring/api/v3>), customers can have full visibility into the applications that are running on Cloud Monitoring (<https://cloud.google.com/monitoring>).

To store documents for each cable system, the company uses Google Drive (https://www.google.com/intl/en_sg/drive/). "We have also done some work with building management systems, and we use Pub/Sub (<https://cloud.google.com/pubsub>) for publishing telemetry data, and to transfer data where needed," says Dr. Madanagopal.

For security purposes, every customer of NMSWorks is required to have a Google account, so they are able to keep all data and information within the same network. But beyond that, the company leverages Google Cloud Armor (<https://cloud.google.com/armor>) for an added layer of security against possible attacks. In addition, when using Cloud Load Balancing, NMSWorks is able to segregate its back-end servers so that they only have local VPCs, instead of global access.

Preparing for the future of 5G networks with TensorFlow

Apart from development and field testing that runs on premises in its labs, all of NMSWorks projects are now run in the cloud. "We've never seen downtime with Google Cloud. The support team is also fantastic, they attend to us almost immediately

every time we run into any issues, and provide good guidance and handholding," says Dr. Madanagopal.

NMSWorks is already running machine learning predictions to prepare for the high resource usage of 5G with TensorFlow (<https://www.tensorflow.org/>).

"We're running time series prediction and resource usage prediction, where we would be able to use that data to do some configuration on our backend to avoid service disruption," explains Dr. Madanagopal.

Currently, the team is working with synthetic data, but are already seeing some promising results. "With real data, we will continue to use the capabilities of Google Cloud to run predictions that can help further fine-tune our products for future 5G projects. This is our future product development roadmap," he concludes.

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